Feature importance on car pricing

1.0 Business problem

Car dealerships, automotive businesses, and consumers need to accurately determine vehicle prices based on car features to make data-driven decisions about surrounding car pricing.

The price of a car is affected by many factors; different cars cater to different groups, targeting wealthy or mainstream consumers. We aim to understand the different factors that drive the different pricing by analysing the impact of different attributes on our feature variable, price. We aim to understand the key features that stand out to have a relationship with price and utilise them to create a model that can accurately predict price.

A price prediction model has significant value beyond simply helping dealerships assign prices. It can work to support customers in looking for cars within their budget or filter through features. Also, it can support the wider stakeholders across the car industry by enabling data-driven decision-making.

2.0 Data preprocessing

Attribute name	**Missing value**	**outlier**	**Data error**
symboling			
normalised_losses	74	yes	
make			2
fuel_type			
aspiration			
num_of_doors	3		
body_style			
drive_wheels			
engine_location			
wheel_base		yes	
length		yes	
width		yes	
height			
curb_weight			
engine_type			

num_of_cylinders			
engine_size		yes	
fuel_system			
bore	4		
stroke	4	yes	
Compression_ratio		yes	
horsepower	2	yes	
peak_rpm	2	yes	
city_mpg		yes	
highway_mpg		yes	
price	4	yes	

We found 93 missing values within the dataset, 74 of which belong to normalised losses. 93 is far too much to simply omit; thus, it was decided to remove the attribute normalised losses, which contained the most missing values, to retain as many rows as possible. With attributes peak_rpm, stroke and bore, which add up to a collective of 14 missing values, we decided to replace them with the mean. For num_of_doors, the mean would not give a whole number, so we used the mode instead. For the price, which is our feature variable, we decided to remove the 4 missing values due to two reasons: one being the small number of missing values, and secondly, we don't want to keep our feature attribute as pure as possible. Lastly, horsepower was cleaned using the median due to the right skewness of the data. After cleaning up missing values, we are left with 401 rows and 25 columns.

Here, we decided to use the lower and upper fence methods to determine what an outlier is. We built a function that allows us to remove the outliers for selected attributes. The table above shows the outliers found using boxplots. However, removing all the outliers would impact our data too much, thus it was decided to clean the feature attribute price and only the other attributes used further in the report, including length, width, engine_size and horsepower. After deletion of the outliers, we were then left with 361 rows and 25 columns.

For data errors, we uncovered two spelling mistakes in the car 'make'. This is with Alfa Romeo, which should instead be spelled alfa-romeo and peugot which should be spelled peugeot.

Our final dataset size post cleaning is 361 rows and 25 columns. This means that from the original 405 rows, we have retained 89.14%. Figure 2.1

3.0 Visualisation

Figure 3.1
This box plot shows the distribution of prices for each car make. Highlighted in different colours, we have made a different box plot for each make. Analysing this, we see three different groups with cars that overall rest below 15000, at 15000 and above 15000 dollars. Higher-priced car makes include Volvo, Porsche, Mercedes-Benz and BMW. With this, we also see car makes with a wider distribution. This could correlate to how many data points we have for the car make, and also for how diverse the car make is with its different pricing of cars. Volvo, Toyota, Nissan, and Mazda are big examples. Overall, the different box plots show a key price range that each car make tries to fit into, wither higher end luxury makes costing more and vice versa.
Figure 3.2
The facit scatter plot here also reinforces the focus that different car makes have, where we have plotted the number of cylinders to be shown with different colours. We see makes such as BMW, Audi, and Mercedes-Benz with blue and red points indicating a 5 or 6-cylinder car, a higher quality and more powerful engine. It is also noted that there is a trend of a higher number of cylinders corresponding to a higher horsepower. A good example is looking at Nissan, which has a clear separation of 4 and 6-cylinder cars forming 2 clusters. The blue cluster sits at a higher horsepower and higher price, while the green cluster is lower on both attributes. The linear trend line that was added to each plot highlights a slight positive trend, indicating that with a higher horsepower, we also have a higher price. This fscit plot shows a multitude of things, but in summary, higher end makes products engine switch more horsepower and number of cylinders, making them more valuable and higher priced.
Figure 3.3
The scatterplot shown here looks at the effect of car size on price. We have also separated based on the car_type. As a general trend seen on all graphs, we see that lighter points, indicating a higher price, have a relationship with car length. This is exemplified clearly with sedan cars, where car lengths between 180 and 200 are priced higher than those of 18 and 160. However, looking at the effect of height, we can't say the same. As shown by hatchbacks and wagons, the height of a car does not equal a higher-priced car; rather, it shows little effect, where lower and higher priced cars will have both low and high heights
Figure 3.4
Similar to size, we have weight. Using a similar chart as before, we displayed the relationship between curb weight and engine size. Overall, the plots indicate a positive relationship, with a higher curb weight having a higher engine size. The higher-priced cars also have a relationship with higher curb weight and engine size. It is good to note that convertibles and hardtops have hardly any points, and it is difficult to make a clear analysis. Looking at figures 3.3 and 3.4, we see that the different car types don't have any clear correlation with pricing, where each make has a mix of high and low-priced cars. Sedans do have some of the lighter point sindicating higher priced cars, but they also have just as many of the dark points.

4.0 Predictive model

- Explore data

After analysis of the visualisation above, we decided to use three different attributes to build a Scatterplot. The attributes are car make, horsepower and curb weight. These three attributes focus on different areas that affect the price of a car. Car makes surround the quality, features

and who the car is targeted for, whether it be for luxury or common use. Horsepower focuses on the engine and the car's capabilities. Curb weight reflects the size and materials the car is composed of. The aim was to avoid using similar attributes to avoid multicollinearity, which simply put is attributes that give the same information.
Chart 4.1 depicts a scatterplot of horsepower and price, showing a clear positive linear relationship where the higher the horsepower, the higher the price. This suggests that more powerful vehicles tend to be more expensive, likely due to better performance and higher-end specifications. The linear line also shows a clear positive trend and enhances the linearity of the data. Chart 4. 2 Similarly, the scatterplot of curb_weight and price does the same, also showing signs of positive linearity with higher curb weight and higher price. builds on Chart 4.1 by adding car make as a categorical variable represented by colour, which allows for further analysis of how different brands cluster in terms of horsepower and price.

- Evaluate

We have tested several different models that led to our final mode testing different attributes and different combinations. One variation was curb_weight and horsepower using two numerical attributes, which we know have a high correlation with price; this created a strong model with an R-squared value of 0.7374. We also tested horsepower by itself; however, the results were weak to moderate, with an R-squared value of 0.514. This could show that simple linear regression with one attribute is weaker in performance. We also looked at other attributes which we have not covered in this report, including a combination of city_mpg and highway_mpg, but the linear relationship with price was not high, giving a R-squared value of 0.5164. Similar to the final model, we tested curb_weight, horsepower and body_style and came up with a strong R-squared value of 0.7666; however, this fell a bit behind our final model. With this, we found that curb weight and horsepower together give a strong model, but it can be further enhanced by adding a third, such as make and body style. Another test was using a fourth attribute, curb_weight, horsepower, length and height, which gave a strong value of 0.7427, but the margin of improvement for the added variables was minor. The trade-off with model simplicity by adding too many variables could also take away from the model's effective use.

- create dummy variables

We have created dummy variables, making it so that the model reads each car make as a numerical value. One per brand minus a baseline, this simply allows the model to read categorical values.

- Fit the model

We ended up using a multiple linear regression model
y = intercept + curbwieght * x1 + horsepower * x2 + car make * x3
Equation 4.6

- Interpret

The performance of the model is very strong with an R-squared value of 0.8555 and an overall p-value of 2.2e-16, a very small number. The p-value of curb weight is 2e-16, and horsepower is 3.04e-16, making them strong indicators and predictors of price. The different car makes also have a low p-value, the highest being Mercedes-Benz at 4.63e-05, though most models sat between 1 and 0. The full list is in Figure 4.4. To further evaluate the model, we made a residual plot shown in Figure 4.5, and it shows that a lot of the points are clustered around the central line, though there are some that sit higher and lower.

- Use the model

Given an example of Curb weight = 2500, Horsepower = 150, Make = BMW, we can fit this into the equation (4.6) $price=-4353.445+5.250\times2500+44.561\times150+5210.724$, which predicts that the price for the car based on the above features is $20,666.429. Attached is a dataset, df1, containing the actual, predicted and residuals made using the model.

- Explain model

The multiple linear regression model estimates car price based on curb weight, horsepower, and car make. Car make is included as a categorical variable using encoded variables, with one make serving as a reference category. The coefficients for each make represent the difference in price according to different baselines. For example, BMW and Mercedes-Benz have large positive coefficients, thus have higher prices compared to the baseline, while brands such as Chevrolet and Honda have negative coefficients, indicating lower prices.

- Key findings

Overall, all three attributes selected help to make a strong linear model that is able to predict price. The high R-squared value of 0.8554 reflects this statement. There is a strong positive linear relationship between attributes horsepower and curb weight on price, making them good predictors for the model. Car brands, though categorical in nature, are also good predictors, showing separating price based on different brands. With the coefficients given for each car make, we can also determine the impact different car brands have on price. Volvo, Porsche, Mercedes-Benz, Audi and BMW indicate a higher price, while the others indicate a lower-priced car. Another key finding is that with the residual plot, we see that lower-priced cars are more centred around the central line, whereas higher-priced cars stray further. This is due to more data being supplied for lower-priced cars compared to higher-priced ones; the model is better at predicting lower-priced cars. In short, the model has been trained and has seen more cars at a lower price.

It was also found that a log transformation of price can boost the performance of the model. The residual plot shown in Figure 4.5 shows increasing variance as fitted values increase. This is partly due to fewer data points, as stated before, but also indicates heteroscedasticity. This suggests the linear model may not fully capture variability in higher-priced cars. A log transformation on the attribute price can reduce variance and improve model fit, making the residuals closer to the centre line, especially at the upper end of the price range, which had the wider variance.

5.0 Recommendations

- Horsepower and curb weight have the greatest impact on price.

Figure 3. 2 and 3.4 work to support this claim with the different facit plots showing a positive linear relationship with price. This is where the curb weight and horsepower of the car increase, and so does the price. With this, stakeholders can assume that they would need to pay or charge more for those higher numbers in those two attributes. The scatterplots made to explore the regression model also further prove the point where a linear line is drawn, and the points are clustered around the line, justifying our model, which uses the two attributes as independent variables along with make.

- Categorical variables also play a role in predicting price

When testing the machine learning model, body style and make were both tested as a third attribute for our multiple regression model. Both increased the R-squared value of the model, showing that they can make great independent variables. Though it was notable that the car make made the best ctaegoircl variable given the result in a high R-squared value of 0.8555. One recommendation is to record further data for different car makes/ brands as some had very few points to make solid conclusions off and given how it already has such an impact a deeper dataset which includes more of the higher end car makes in particular and enhance future predcitbly of the model we created, figure 3.1 hiuglights this with a very small range of certain makes including mercedes benz and mercury. Also, other categorical features that describe a car could be measured. If we recall all our vidation made, we see that conclusions can be made from them all.

- Price is predictive based on the different characteristics of cars

Many of the different attributes, when plotted on a scatterplot with a linear line, show a positive linear relationship with price. Figure 3.2,3.4 and figures 4.1 to 4.3 all reflect a degree of linearity with price. This works to prove thatprice if cars is not random and that there is a reason behind why dealerships price their cars as is. Dealerships and buyers should rely on key measurable attributes when evaluating how a car should be priced. By comparing a vehicle's features against expected price trends by means such as our model, businesses can set fair prices, while consumers can better identify where a car has been appropriately priced. The predictability can also be improved by improving the model. Further tests have shown that by transforming the price attribute with a log transformation, the model can improve results, shown with a 0.8611 R-squared value.

6.0 conclusion

The business problem of helping car dealerships, automotive businesses and consumers understand features that contribute to car pricing and to help them calculate price using our multiple regression model was achieved. This was done through visualisations 3.1 to 3.4, which helped understand car make, body styles, horsepower, curbweight, length and height and the impact it has on car pricing. Also, through our independent variables, curbweight, horsepower, and make to make a predictive model to accurately predict price. This report has successfully shown that price is not random but rather predictive based on a car's features.

7.0 Reference

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8.0 Appendix

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<tr><td>1.0</td><td>Business problem</td><td></td></tr>
<tr><td>2.0</td><td>Data preprocessing</td><td>2.1
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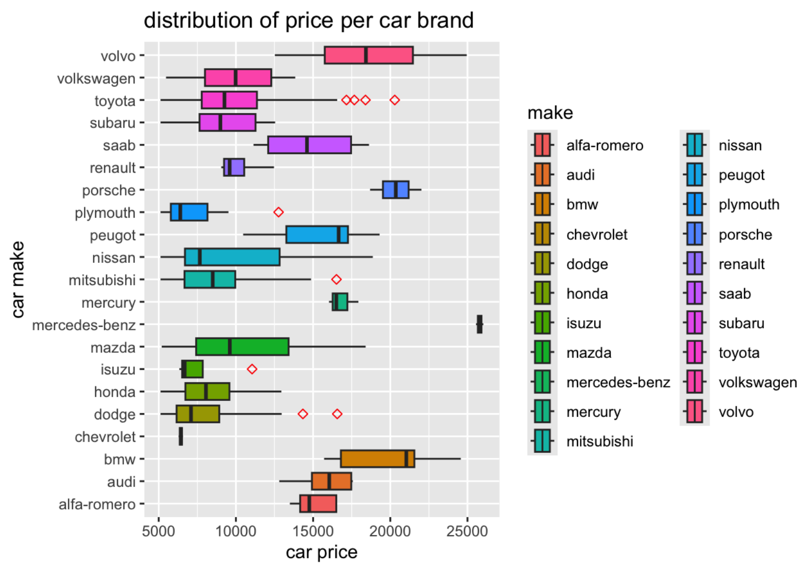

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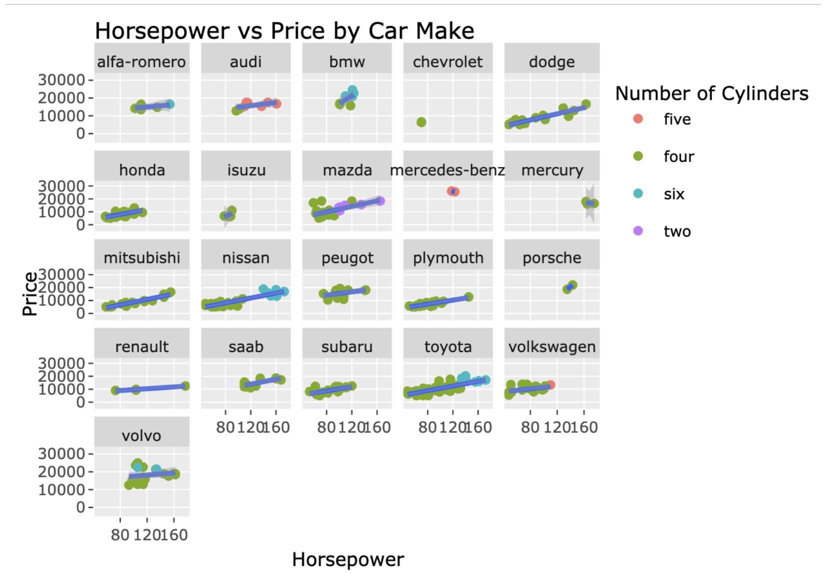

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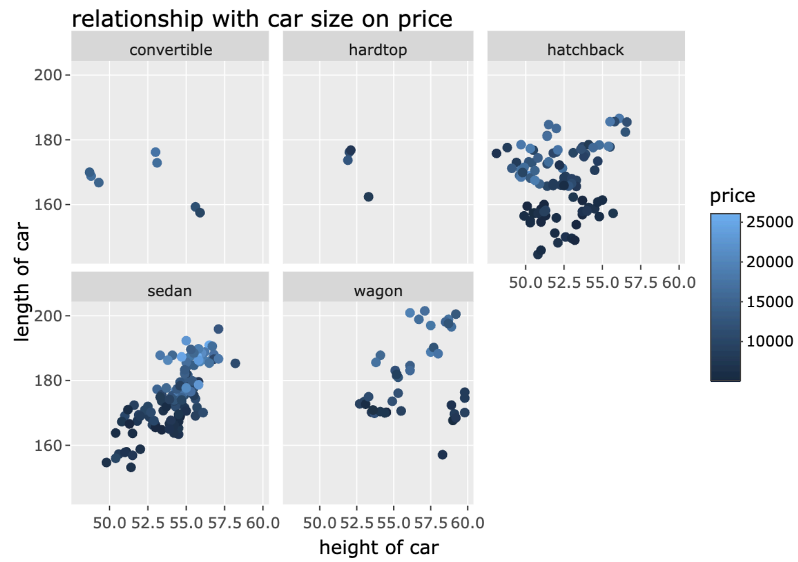
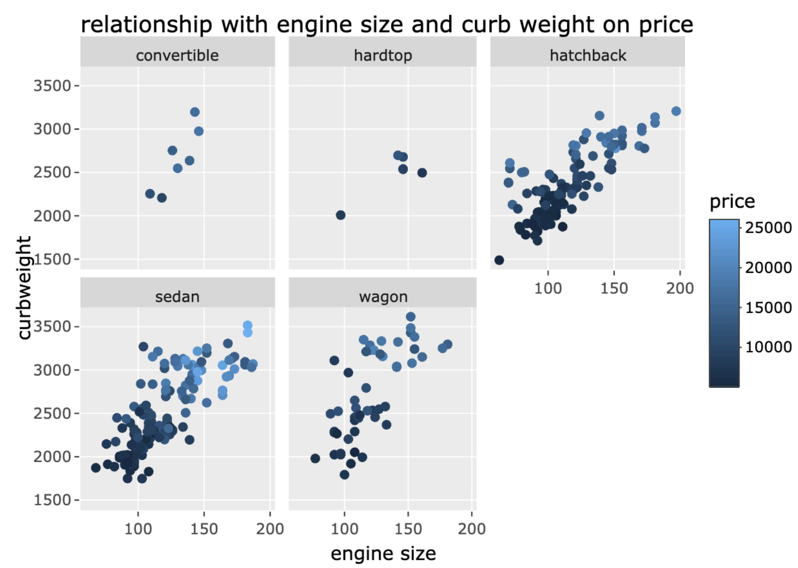
		 3.4 
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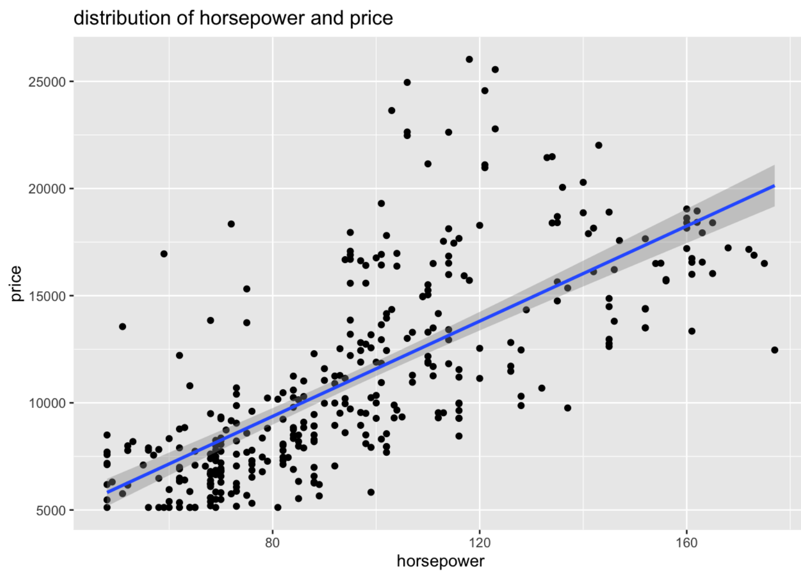
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horsepower
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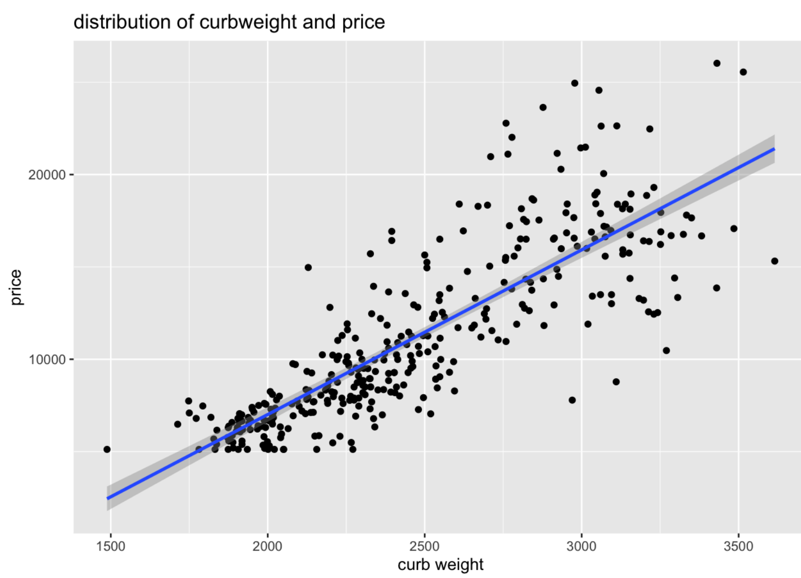
15000
10000
5000
80
120
160

4.2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curb weight
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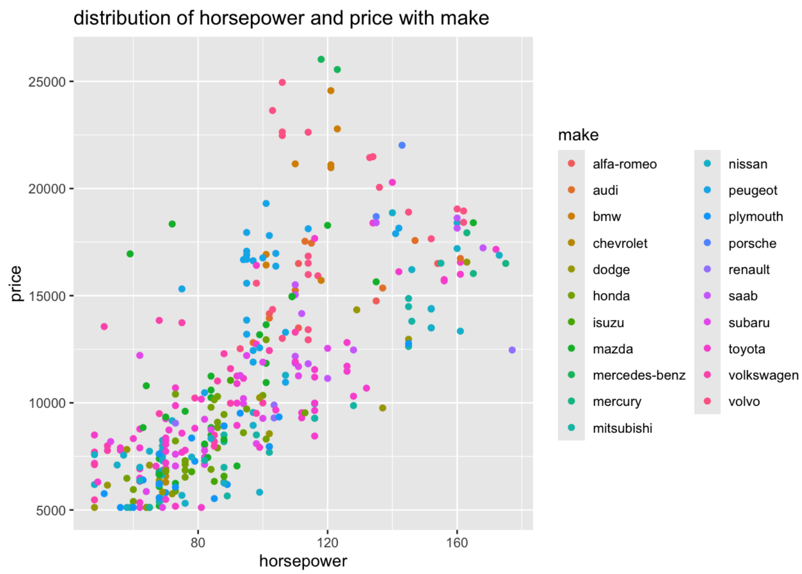
10000
1500
2000
2500
3000
3500

4.3



4.4

```
Coefficients:
                     Estimate Std. Error t value Pr(>|t|)    
(Intercept)        -4353.4447  1213.5727  -3.587 0.000383 ***
curb_weight            5.2496     0.4503  11.659  < 2e-16 ***
horsepower            44.5614     5.1785   8.605 2.92e-16 ***
makeaudi             635.9436  1049.8719   0.606 0.545098    
makebmw             5210.7240  1050.7294   4.959 1.12e-06 ***
makechevrolet      -2260.3882  1561.6033  -1.447 0.148690    
makedodge          -2535.3170   932.6274  -2.718 0.006897 ** 
makehonda          -2225.5941   913.1909  -2.437 0.015318 *  
makeisuzu          -3962.0073  1243.4611  -3.186 0.001576 ** 
makemazda           -953.2852   898.2361  -1.061 0.289318    
makemercedes-benz  6541.0882  1585.2204   4.126 4.65e-05 ***
makemercury        -1443.4830  1356.8255  -1.064 0.288148    
makemitsubishi     -3459.6794   922.0006  -3.752 0.000206 ***
makenissan         -2557.8034   884.1904  -2.893 0.004066 ** 
makepeugeot        -1517.1190   966.0218  -1.570 0.117239    
makeplymouth       -3177.0927   943.8413  -3.366 0.000850 ***
makeporsche         3763.1593  1541.9976   2.440 0.015182 *  
makerenault        -3432.8938  1236.7815  -2.776 0.005816 ** 
makesaab           -1143.1709   980.7944  -1.166 0.244616    
makesubaru         -2614.7465   922.4877  -2.834 0.004866 ** 
maketoyota         -2616.4476   862.7003  -3.033 0.002610 ** 
makevolkswagen     -1265.0628   932.8108  -1.356 0.175946    
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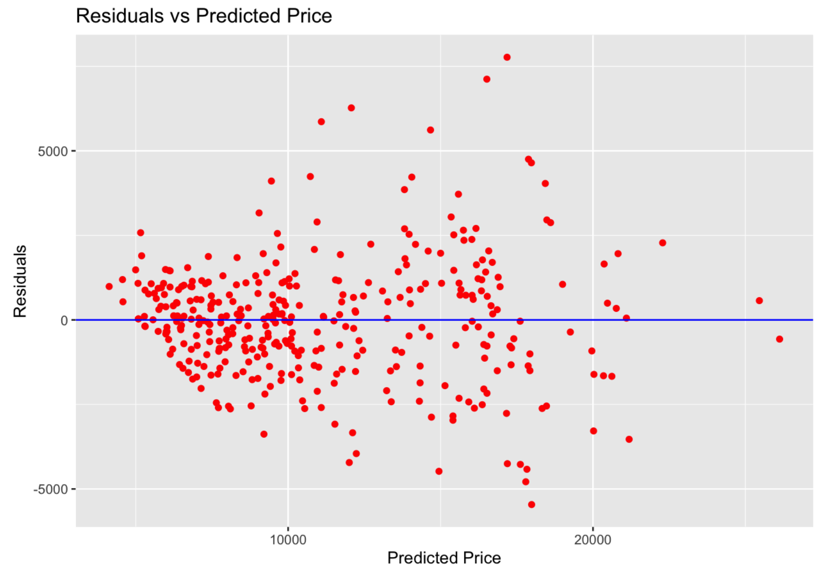
4.5

		 4.6 $\text{price} = -4353.445 + 5.250 \cdot \text{curb\_weight} + 44.561 \cdot \text{horsepower}$ $+ 635.944 \cdot \text{makeAudi}$ $+ 5210.724 \cdot \text{makeBMW}$ $- 2260.388 \cdot \text{makeChevrolet}$ $- 2535.317 \cdot \text{makeDodge}$ $- 2225.594 \cdot \text{makeHonda}$ $- 3962.007 \cdot \text{makeIsuzu}$ $- 953.285 \cdot \text{makeMazda}$ $+ 6541.088 \cdot \text{makeMercedesBenz}$ $- 1443.483 \cdot \text{makeMercury}$ $- 3459.679 \cdot \text{makeMitsubishi}$ $- 2557.803 \cdot \text{makeNissan}$ $- 1517.119 \cdot \text{makePeugeot}$ $- 3177.093 \cdot \text{makePlymouth}$ $+ 3763.159 \cdot \text{makePorsche}$ $- 3432.894 \cdot \text{makeRenault}$ $- 1143.171 \cdot \text{makeSaab}$ $- 2614.747 \cdot \text{makeSubaru}$ $- 2616.448 \cdot \text{makeToyota}$ $- 1265.063 \cdot \text{makeVolkswagen}$ $+ 1179.240 \cdot \text{makeVolvo}$
5.0	recommendations	
6.0	conclusion	